INCH-POUND

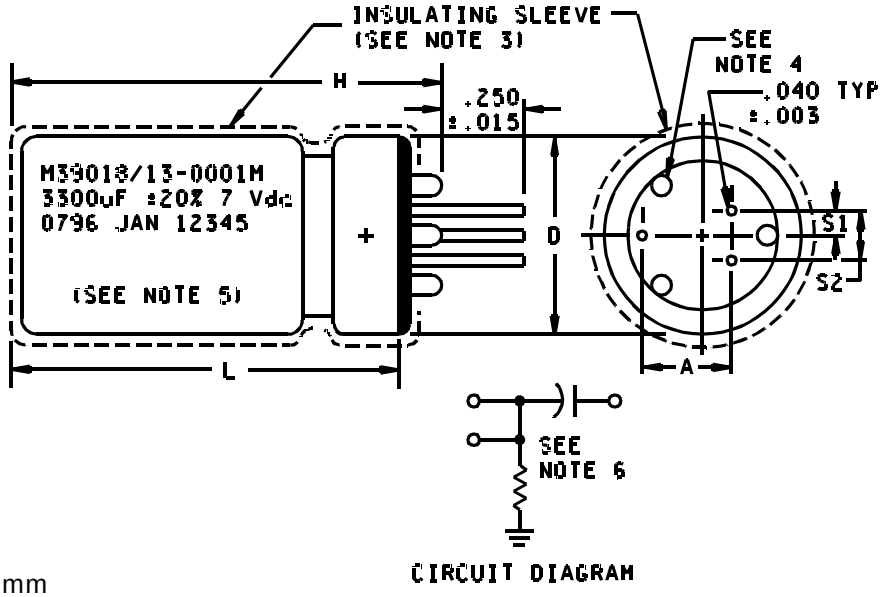
MIL-PRF-39018/13
11 October 1996

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, ELECTROLYTIC (ALUMINUM OXIDE)(POLARIZED),
ESTABLISHED RELIABILITY, STYLE CUR22 (INSULATED)

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-39018.



Inches	mm
.003	0.08
.015	0.38
.040	1.02
.250	6.35

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Case size	Dimensions					
	L ± .062 (1.57)	D ± .015 (0.38)	H max	A ± .015 (0.38)	S1 ± .015 (0.38)	S2 ± .015 (0.38)
A0	1.125 (28.58)	.750 (19.05)	1.246 (31.65)	.300 (7.62)	.100 (2.54)	.200 (5.08)
B0	1.625 (41.28)	.750 (19.05)	1.746 (44.35)	.300 (7.62)	.100 (2.54)	.200 (5.08)
C0	2.125 (53.98)	.750 (19.05)	2.246 (57.05)	.300 (7.62)	.100 (2.54)	.200 (5.08)
D0	2.625 (66.68)	.750 (19.05)	2.746 (69.75)	.300 (7.62)	.100 (2.54)	.200 (5.08)
E0	3.125 (79.38)	.750 (19.05)	3.246 (82.45)	.300 (7.62)	.100 (2.54)	.200 (5.08)
F0	3.625 (92.08)	.750 (19.05)	3.746 (95.15)	.300 (7.62)	.100 (2.54)	.200 (5.08)
A1	1.125 (28.58)	.875 (22.22)	1.246 (31.65)	.400 (10.16)	.150 (3.81)	.300 (7.62)
B1	1.625 (41.28)	.875 (22.22)	1.746 (44.35)	.400 (10.16)	.150 (3.81)	.300 (7.62)
C1	2.125 (53.98)	.875 (22.22)	2.246 (57.05)	.400 (10.16)	.150 (3.81)	.300 (7.62)
D1	2.625 (66.68)	.875 (22.22)	2.746 (69.75)	.400 (10.16)	.150 (3.81)	.300 (7.62)
E1	3.125 (79.38)	.875 (22.22)	3.246 (82.45)	.400 (10.16)	.150 (3.81)	.300 (7.62)
F1	3.625 (92.08)	.875 (22.22)	3.746 (95.15)	.400 (10.16)	.150 (3.81)	.300 (7.62)
A2	1.125 (28.58)	1.000 (25.40)	1.246 (31.65)	.400 (10.16)	.150 (3.81)	.300 (7.62)
B2	1.625 (41.28)	1.000 (25.40)	1.746 (44.35)	.400 (10.16)	.150 (3.81)	.300 (7.62)
C2	2.125 (53.98)	1.000 (25.40)	2.246 (57.05)	.400 (10.16)	.150 (3.81)	.300 (7.62)
D2	2.625 (66.68)	1.000 (25.40)	2.746 (69.75)	.400 (10.16)	.150 (3.81)	.300 (7.62)
E2	3.125 (79.38)	1.000 (25.40)	3.246 (82.45)	.400 (10.16)	.150 (3.81)	.300 (7.62)
F2	3.625 (92.08)	1.000 (25.40)	3.746 (95.15)	.400 (10.16)	.150 (3.81)	.300 (7.62)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. Insulating sleeve thickness shall not exceed .008 (0.20 mm) and shall overlap the ends of the capacitor body.
4. Standoff shape is optional. The minimum height of the standoffs shall be .025 (0.64 mm).
5. Example of marking and its relative location on the capacitor body.
6. There is an indeterminate resistance between the metal case and the negative terminal.

FIGURE 1. Dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Tubular, metal.

Terminals: See figure 1.

DC rated voltage: See table I.

Reverse voltage: 1.5 volts maximum.

Capacitance (Cap.): See table I.

Capacitance tolerance: ±20 percent.

Failure rate level: M, P, R, and S.

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TABLE I. Style CUR22 capacitors.

Capacitance μF	Maximum ESR at +25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ac ripple current at +85°C 20 kHz - 100 kHz A	Case size	Dash number 1/
	120 Hz ohms	20 kHz ohms	Low temperature ohms	+25°C ohms	+25°C μA	+105°C μA			
7 volts dc rated voltage at +85°C and +105°C 10 volts dc surge voltage at +85°C and +105°C									
3300	0.074	0.049	1.175	0.047	76	456	3.46	A0	0001-
4700	0.050	0.031	0.775	0.031	91	544	4.55	A1	0002-
6800	0.033	0.019	0.550	0.022	109	655	6.09	C0	0003-
6800	0.033	0.016	0.450	0.018	109	655	6.60	B1	0004-
6800	0.035	0.027	0.675	0.027	109	655	5.06	A2	0005-
10,000	0.022	0.015	0.450	0.018	132	794	7.40	D0	0006-
10,000	0.021	0.012	0.375	0.015	132	794	8.05	C1	0007-
10,000	0.024	0.014	0.425	0.017	132	794	7.60	B2	0008-
15,000	0.019	0.012	0.375	0.015	162	972	9.06	E0	0009-
15,000	0.016	0.009	0.300	0.012	162	972	9.89	D1	0010-
15,000	0.017	0.011	0.375	0.015	162	972	9.05	C2	0011-
18,000	0.017	0.010	0.350	0.014	177	1065	10.10	F0	0012-
22,000	0.011	0.007	0.275	0.011	196	1177	11.94	E1	0013-
22,000	0.012	0.008	0.325	0.013	196	1177	11.09	D2	0014-
27,000	0.010	0.006	0.250	0.010	217	1304	13.40	F1	0015-
27,000	0.011	0.007	0.300	0.012	217	1304	13.59	E2	0016-
33,000	0.010	0.006	0.275	0.011	240	1442	14.92	F2	0017-
10 volts dc rated voltage at +85°C and +105°C 15 volts dc surge voltage at +85°C and +105°C									
2200	0.079	0.050	1.150	0.046	74	445	3.45	A0	0018-
3300	0.054	0.031	0.775	0.031	91	545	4.53	A1	0019-
3900	0.045	0.026	0.675	0.027	99	592	5.12	B0	0020-
4700	0.039	0.027	0.675	0.027	108	650	5.04	A2	0021-
5600	0.033	0.019	0.550	0.022	118	710	6.06	C0	0022-
5600	0.027	0.016	0.450	0.018	118	710	6.58	B1	0023-
6800	0.030	0.014	0.425	0.017	130	782	7.40	B2	0024-
8200	0.022	0.015	0.450	0.018	143	859	7.36	D0	0025-
10,000	0.015	0.009	0.300	0.012	158	949	9.84	D1	0026-
10,000	0.017	0.011	0.375	0.015	158	949	9.01	C2	0027-
12,000	0.016	0.010	0.325	0.013	173	1039	8.83	F0	0028-
15,000	0.012	0.007	0.250	0.010	194	1162	11.88	E1	0029-
18,000	0.010	0.006	0.200	0.008	212	1273	13.90	F1	0030-
22,000	0.010	0.006	0.275	0.011	235	1407	14.85	F2	0031-

See footnote at end of table.

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TABLE I. Style CUR22 capacitors - Continued.

Capacitance μF	Maximum ESR at +25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ac ripple current at +85°C 20 kHz - 100kHz A	Case size	Dash number 1/
	120 Hz ohms	20 kHz ohms	Low temperature ohms	+25°C ohms	+25°C μA	+105°C μA			
16 volts dc rated voltage at +85°C and +105°C 20 volts dc surge voltage at +85°C and +105°C									
1500	0.097	0.050	1.175	0.047	77	465	3.43	A0	0032-
2200	0.066	0.026	0.675	0.027	94	563	5.10	B0	0033-
2200	0.072	0.031	0.775	0.031	94	563	4.50	A1	0034-
2700	0.058	0.027	0.700	0.028	104	624	5.01	A2	0035-
3300	0.044	0.020	0.550	0.022	115	689	6.03	C0	0036-
3300	0.048	0.016	0.450	0.018	115	689	6.55	B1	0037-
4700	0.028	0.015	0.450	0.018	137	823	7.33	D0	0038-
4700	0.031	0.012	0.375	0.015	137	823	7.97	C1	0039-
4700	0.038	0.014	0.425	0.017	137	823	7.37	B2	0040-
6800	0.026	0.012	0.375	0.015	165	990	8.98	E0	0041-
6800	0.024	0.009	0.300	0.012	165	990	9.79	D1	0042-
6800	0.026	0.011	0.375	0.015	165	990	8.97	C2	0043-
8200	0.024	0.009	0.350	0.014	181	1087	10.30	F0	0044-
10,000	0.016	0.007	0.275	0.011	200	1200	11.83	E1	0045-
10,000	0.017	0.008	0.325	0.013	200	1200	10.98	D2	0046-
12,000	0.014	0.006	0.225	0.009	219	1315	13.30	F1	0047-
12,000	0.015	0.007	0.300	0.012	219	1315	13.46	E2	0048-
15,000	0.010	0.006	0.275	0.011	245	1470	14.78	F2	0049-
30 volts dc rated voltage at +85°C and +105°C 40 volts dc surge voltage at +85°C and +105°C									
680	0.130	0.051	1.200	0.048	71	428	3.41	A0	0050-
820	0.111	0.032	0.800	0.032	78	471	4.46	A1	0051-
1000	0.088	0.027	0.700	0.028	87	520	5.05	B0	0052-
1200	0.078	0.028	0.700	0.028	95	569	4.97	A2	0053-
1500	0.049	0.020	0.575	0.023	106	636	5.97	C0	0054-
1500	0.062	0.017	0.475	0.019	106	636	6.49	B1	0055-
2200	0.039	0.015	0.475	0.019	128	771	7.27	D0	0056-
2200	0.042	0.013	0.375	0.015	128	771	7.90	C1	0057-
2200	0.044	0.014	0.450	0.018	128	771	7.31	B2	0058-
3300	0.032	0.012	0.400	0.016	157	944	8.90	E0	0059-
3300	0.026	0.010	0.325	0.013	157	944	9.72	D1	0060-
3300	0.029	0.011	0.400	0.016	157	944	8.90	C2	0061-
3900	0.031	0.011	0.375	0.015	171	1026	10.00	F0	0062-
3900	0.025	0.008	0.275	0.011	171	1026	11.74	E1	0063-
3900	0.025	0.009	0.325	0.013	171	1026	10.88	D2	0064-
4700	0.023	0.007	0.250	0.010	188	1126	13.10	F1	0065-
5600	0.017	0.007	0.300	0.012	205	1230	13.36	E2	0066-
6800	0.013	0.006	0.300	0.012	226	1355	14.68	F2	0067-

See footnote at end of table.

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TABLE I. Style CUR22 capacitors - Continued.

Capacitance μF	Maximum ESR at +25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ac ripple current at +85°C 20 kHz - 100kHz A	Case size	Dash number 1/
	120 Hz ohms	20 kHz ohms	Low temperature ohms	+25°C ohms	+25°C μA	+105°C μA			
50 volts dc rated voltage at +85°C and +105°C 75 volts dc surge voltage at +85°C and +105°C									
390	0.237	0.126	4.200	0.120	70	419	2.16	A0	0068-
560	0.166	0.066	2.205	0.063	84	502	3.21	B0	0069-
560	0.173	0.083	2.730	0.078	84	502	2.76	A1	0070-
680	0.146	0.077	2.520	0.072	92	553	3.00	A2	0071-
820	0.113	0.049	1.680	0.048	101	607	3.80	C0	0072-
820	0.118	0.043	1.435	0.041	101	607	4.01	B1	0073-
1200	0.077	0.037	1.295	0.037	122	735	4.62	D0	0074-
1200	0.081	0.032	1.085	0.031	122	735	4.88	C1	0075-
1200	0.083	0.040	1.365	0.039	122	735	4.41	B2	0076-
1500	0.062	0.029	1.015	0.029	137	822	5.66	E0	0077-
1500	0.066	0.030	1.085	0.031	137	822	5.37	C2	0078-
1800	0.056	0.026	0.910	0.026	150	900	6.35	F0	0079-
1800	0.054	0.025	0.840	0.024	150	900	6.00	D1	0080-
2200	0.044	0.019	0.665	0.019	166	995	7.24	E1	0081-
2200	0.045	0.023	0.840	0.024	166	995	6.57	D2	0082-
2700	0.040	0.017	0.595	0.017	184	1102	8.12	F1	0083-
3300	0.030	0.018	0.700	0.020	203	1219	8.05	E2	0084-
3900	0.025	0.016	0.630	0.018	221	1325	8.84	F2	0085-
63 volts dc rated voltage at +85°C and +105°C 84 volts dc surge voltage at +85°C and +105°C									
220	0.289	0.130	4.235	0.121	59	353	2.13	A0	0086-
330	0.201	0.086	2.765	0.079	72	433	2.72	A1	0087-
390	0.163	0.068	2.240	0.064	78	470	3.16	B0	0088-
470	0.146	0.079	2.590	0.074	86	516	2.96	A2	0089-
560	0.114	0.051	1.715	0.049	94	563	3.74	C0	0090-
560	0.118	0.045	1.470	0.042	94	563	3.95	B1	0091-
680	0.104	0.041	1.400	0.040	103	621	4.34	B2	0092-
820	0.077	0.039	1.330	0.038	114	682	4.55	D0	0093-
820	0.081	0.033	1.120	0.032	114	682	4.81	C1	0094-
1000	0.064	0.030	1.050	0.030	125	753	5.57	E0	0095-
1000	0.071	0.031	1.120	0.032	125	753	5.29	C2	0096-
1200	0.057	0.027	0.945	0.027	137	825	6.25	F0	0097-
1200	0.055	0.025	0.875	0.025	137	825	5.91	D1	0098-
1500	0.044	0.020	0.665	0.019	154	922	7.14	E1	0099-
1500	0.047	0.024	0.875	0.025	154	922	6.47	D2	0100-
1800	0.040	0.018	0.595	0.017	168	1010	8.02	F1	0101-
1800	0.039	0.018	0.700	0.020	168	1010	7.93	E2	0102-
2200	0.032	0.016	0.665	0.019	186	1117	8.71	F2	0103-

See footnote at end of table.

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TABLE I. Style CUR22 capacitors - Continued.

Capacitance μF	Maximum ESR at +25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ac ripple current at +85°C 20 kHz-100kHz A	Case size	Dash number 1/
	120 Hz ohms	20 kHz ohms	Low temperature ohms	+25°C ohms	+25°C μA	+105°C μA			
75 volts dc rated voltage at +85°C and +105°C 100 volts dc surge voltage at +85°C and +105°C									
150	0.424	0.134	4.305	0.123	53	318	2.10	A0	0104-
220	0.289	0.070	2.275	0.065	64	385	3.12	B0	0105-
220	0.302	0.089	2.835	0.081	64	385	2.68	A1	0106-
270	0.222	0.082	2.660	0.076	71	427	2.91	A2	0107-
330	0.193	0.052	1.715	0.049	79	472	3.69	C0	0108-
330	0.201	0.046	1.505	0.043	79	472	3.90	B1	0109-
470	0.135	0.040	1.330	0.038	94	563	4.49	D0	0110-
470	0.141	0.035	1.120	0.032	94	563	4.74	C1	0111-
470	0.150	0.043	1.470	0.042	94	563	4.28	B2	0112-
680	0.114	0.031	1.050	0.030	113	677	5.50	E0	0113-
680	0.098	0.026	0.875	0.025	113	677	5.82	D1	0114-
680	0.104	0.032	1.155	0.033	113	677	5.21	C2	0115-
820	0.102	0.028	0.945	0.027	124	744	6.11	F0	0116-
820	0.080	0.021	0.700	0.020	124	744	7.03	E1	0117-
820	0.086	0.024	0.910	0.026	124	744	6.38	D2	0118-
1000	0.072	0.019	0.630	0.018	137	822	7.85	F1	0119-
1200	0.059	0.019	0.735	0.021	150	900	7.82	E2	0120-
1500	0.046	0.017	0.700	0.020	168	1006	8.59	F2	0121-
100 volts dc rated voltage at +85°C and +105°C 125 volts dc surge voltage at +85°C and +105°C									
120	0.530	0.137	4.375	0.125	55	329	2.07	A0	0122-
180	0.302	0.091	2.870	0.082	67	402	2.64	A1	0123-
220	0.312	0.072	2.310	0.066	74	445	3.08	B0	0124-
220	0.273	0.084	2.730	0.078	74	445	2.87	A2	0125-
270	0.250	0.054	1.750	0.050	82	493	3.64	C0	0126-
270	0.201	0.047	1.505	0.043	82	493	3.84	B1	0127-
390	0.176	0.041	1.365	0.039	99	592	4.42	D0	0128-
390	0.145	0.036	1.155	0.033	99	592	4.67	C1	0129-
390	0.156	0.044	1.505	0.043	99	592	4.22	B2	0130-
560	0.136	0.032	1.085	0.031	118	710	5.42	E0	0131-
560	0.104	0.027	0.875	0.025	118	710	5.74	D1	0132-
560	0.104	0.033	1.190	0.034	118	710	5.14	C2	0133-
680	0.121	0.031	0.980	0.028	130	782	6.09	F0	0134-
820	0.080	0.021	0.700	0.020	143	859	6.94	E1	0135-
820	0.086	0.025	1.295	0.037	143	859	6.29	D2	0136-
1000	0.071	0.019	0.630	0.018	158	949	7.79	F1	0137-
1000	0.059	0.020	0.770	0.022	158	949	7.71	E2	0138-
1200	0.046	0.017	0.700	0.020	173	1039	8.47	F2	0139-

See footnote at end of table.

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TABLE I. Style CUR22 capacitors - Continued.

Capacitance μF	Maximum ESR at +25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ac ripple current at +85°C 20 kHz - 100kHz A	Case size	Dash number 1/
	120 Hz ohms	20 kHz ohms	Low temperature ohms	+25°C ohms	+25°C μA	+105°C μA			
150 volts dc rated voltage at +85°C and +105°C 175 volts dc surge voltage at +85°C and +105°C									
82	1.230	0.420	14.665	0.419	55	333	0.60	A0	0140-
120	0.884	0.346	10.325	0.295	67	402	0.74	A1	0141-
150	0.670	0.245	7.595	0.217	75	450	0.86	B0	0142-
180	0.560	0.183	5.670	0.162	82	493	0.99	C0	0143-
180	0.589	0.180	5.390	0.154	82	493	0.95	B1	0144-
180	0.589	0.293	9.380	0.268	82	493	0.93	A2	0145-
270	0.380	0.139	4.305	0.123	101	604	1.20	D0	0146-
270	0.393	0.134	4.060	0.116	101	604	1.20	C1	0147-
270	0.393	0.152	4.865	0.139	101	604	1.20	B2	0148-
390	0.300	0.109	3.325	0.095	121	726	1.50	E0	0149-
390	0.272	0.102	3.115	0.089	121	726	1.40	D1	0150-
390	0.272	0.114	3.640	0.104	121	726	1.50	C2	0151-
470	0.270	0.097	2.975	0.085	133	797	1.68	F0	0152-
560	0.189	0.080	2.450	0.070	145	869	2.00	E1	0153-
560	0.189	0.086	2.765	0.079	145	869	1.90	D2	0154-
680	0.168	0.072	2.205	0.063	160	958	2.24	F1	0155-
680	0.156	0.068	2.135	0.061	160	958	2.30	E2	0156-
820	0.129	0.060	1.925	0.055	175	1052	2.60	F2	0157-
200 volts dc rated voltage at +85°C and +105°C 225 volts dc surge voltage at +85°C and +105°C									
68	1.460	0.491	15.260	0.436	58	350	0.55	A0	0158-
100	1.060	0.360	10.745	0.307	71	424	0.67	A1	0159-
120	0.830	0.255	7.910	0.226	77	465	0.78	B0	0160-
120	0.884	0.304	9.765	0.279	77	465	0.76	A2	0161-
150	0.670	0.191	5.880	0.168	87	520	0.90	C0	0162-
150	0.707	0.187	5.600	0.160	87	520	0.87	B1	0163-
220	0.450	0.145	4.445	0.127	105	629	1.10	D0	0164-
220	0.482	0.140	4.235	0.121	105	629	1.10	C1	0165-
220	0.482	0.158	5.075	0.145	105	629	1.10	B2	0166-
270	0.370	0.113	3.465	0.099	116	697	1.30	E0	0167-
330	0.330	0.100	3.080	0.088	128	771	1.45	F0	0168-
330	0.322	0.106	3.220	0.092	128	771	1.40	D1	0169-
330	0.322	0.118	3.780	0.108	128	771	1.40	C2	0170-
390	0.272	0.083	2.555	0.073	140	838	1.60	E1	0171-
470	0.242	0.074	2.275	0.065	153	920	1.79	F1	0172-
470	0.226	0.090	2.870	0.082	153	920	1.80	D2	0173-
560	0.189	0.070	2.240	0.064	167	1004	2.10	E2	0174-
680	0.156	0.063	1.995	0.057	184	1106	2.40	F2	0175-

See footnote at end of table.

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TABLE I. Style CUR22 capacitors - Continued.

Capacitance μF	Maximum ESR at +25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ac ripple current at +85°C 20 kHz - 100kHz A	Case size	Dash number 1/
	120 Hz ohms	20 kHz ohms	Low temperature ohms	+25°C ohms	+25°C μA	+105°C μA			
250 volts dc rated voltage at +85°C and +105°C 275 volts dc surge voltage at +85°C and +105°C									
56	1.710	0.817	26.460	0.756	59	355	0.51	A0	0176-
68	1.560	0.631	19.775	0.565	65	391	0.55	A1	0177-
82	1.160	0.424	13.300	0.380	72	430	0.66	B0	0178-
100	1.060	0.562	15.855	0.453	79	474	0.69	A2	0179-
120	0.800	0.317	9.660	0.276	87	520	0.82	C0	0180-
120	0.884	0.327	9.940	0.284	87	520	0.78	B1	0181-
150	0.707	0.292	8.190	0.234	97	581	0.91	B2	0182-
180	0.530	0.241	7.105	0.203	106	636	1.00	D0	0183-
180	0.589	0.245	7.210	0.206	106	636	1.00	C1	0184-
220	0.430	0.188	5.355	0.153	117	704	1.20	E0	0185-
220	0.482	0.186	5.320	0.152	117	704	1.10	D1	0186-
220	0.482	0.218	6.090	0.174	117	704	1.10	C2	0187-
270	0.390	0.170	4.795	0.137	130	779	1.34	F0	0188-
330	0.322	0.145	4.025	0.115	144	862	1.50	E1	0189-
330	0.322	0.166	4.620	0.132	144	862	1.50	D2	0190-
390	0.290	0.129	3.605	0.103	156	937	1.68	F1	0191-
470	0.226	0.130	3.605	0.103	171	1028	1.90	E2	0192-
560	0.189	0.115	3.185	0.091	187	1122	2.20	F2	0193-

1/ The last digit of the dash number shall indicate the failure rate level symbol ("M", "P", "R", or "S"). For parts with random vibration, the letter R shall be used in place of the dash (i.e., M39018/13R0001M).

REQUIREMENTS - Continued:

Operating temperature range: -55°C to +105°C for voltage ratings of 7 volts through 100 volts; -40°C to +105°C for voltage ratings of 150 volts to 250 volts.

DC leakage (DCL): See table I.

Equivalent series resistance (ESR): See table I.

Low temperature exposure:

DCL - See table I.
Cap. - See table I.
ESR - See table I.

Terminal strength: Method 211 of MIL-STD-202, test condition A (3 pounds) (1.36 kg) and D (3 rotations).

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Stability at reduced and high temperatures:

Step 1 (at 25°C):

DCL - See table I.
Cap. - See table I.
ESR (at 120 Hz) - See table I.

Step 2 (at applicable low temperature):

Impedance - See table I.

ΔCap. - Within the following percentages of the step 1 value:

7 V dc: ±35 percent.
10 V dc through 30 V dc: ±25 percent.
50 V dc through 100 V dc: ±20 percent.
150 V dc through 250 V dc: ±30 percent.

Step 3 (at +25°C):

DCL - See table I.
ΔCap. - Within ±5 percent of step 1 value.
ESR (at 120 Hz) - See table I.

Step 4 (at +105°C):

DCL - See table I.
ΔCap. - Within ±30 percent of step 1 value.

Step 5 (at +25°C):

DCL - See table I.
ΔCap. - Within ±5 percent of step 1 value.
ESR (at 120 Hz) - See table I.

Life: Method 108 of MIL-STD-202.

For qualification: Condition F (2,000 hours) at +105°C.

DCL - See table I.
ΔCap. - Within ±15 percent of initial measured value.
ESR - Not more than 150 percent of initial requirement.

10,000 hour (extended life and group C) at +85°C.

DCL - At +25°C, initial limit; at +85°C, 125 percent of initial limit.
ΔCap. - Within ±25 percent of initial measured value.
ESR - Not more than 200 percent of initial requirement.

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Vibration:

High frequency - Method 204 of MIL-STD-202, test condition A.
Random (when applicable) - In accordance with MIL-PRF-39018.

Thermal shock and immersion:

Thermal shock: Method 107 of MIL-STD-202, condition A, except that in step 1, units shall be tested at the applicable low temperature, and in step 3, units shall be tested at +105°C.

Immersion: Method 104 of MIL-STD-202, condition B (2 cycles, 30 minutes).

DCL - See table I.
ΔCap. - Within ±5 percent of initial measured value.
ESR - See table I.
Case insulation - Shall meet initial requirements.

Surge voltage: In accordance with MIL-PRF-39018.

DCL - See table I.
ΔCap. - Within ±12 percent of initial measured value.
ESR - See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL - See table I.
ΔCap. - Within ±6 percent of initial measured value.
ESR - See table I.
Case insulation - Shall meet initial requirements.

High temperature exposure: In accordance with MIL-PRF-39018.

DCL - Not more than 200 percent of initial requirement.
ΔCap. - Within ±15 percent of initial measured value.
ESR - Not more than 115 percent of initial requirement.

Penetration of solvents: In accordance with MIL-PRF-39018.

High temperature verification: In accordance with MIL-PRF-39018 at +105°C with rated voltage applied.

DCL - See table I.
ΔCap. - Within ±15 percent of initial measured value.
ESR - Not more than 150 percent of initial requirement.

Reverse voltage aging: In accordance with MIL-PRF-39018.

Marking: In accordance with MIL-PRF-39018.

Part or Identifying Number (PIN):M39018/13- (dash number from table I). For parts with random vibration, the letter R shall be used in place of the dash (i.e., M39018/13R0001M).

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APPLICATION NOTES:

Ripple current: Capacitors shall withstand the rms ripple current at 20 kHz through 100 kHz as shown in table I. The maximum ripple current values apply with the following qualifications:

The sum of the applied dc voltage plus the peak value of the impressed ac voltage shall not exceed the dc rated voltage of the capacitor.

The sum of the applied dc voltage plus the peak value of the impressed ac voltage shall be such that not more than 1.5 volt reversal is experienced by the capacitor.

If capacitors are operated at a frequency of less than 20 kHz, the rated 20 kHz rms ripple current shall be multiplied by the factors shown in table II.

TABLE II. Factors for determining ripple current at various frequencies.

Rated voltage V dc	Frequency				
	60 Hz	120 Hz	400 Hz	1 kHz - 19 kHz	20 kHz - 100 kHz
7 through 75	.60	.75	.80	.90	1.0
100 through 250	.43	.54	.75	.85	1.0

Where capacitors are operated at temperatures other than +85°C, the permissible rms ripple current shall be multiplied by the factors in table III.

TABLE III. Factors for determining ripple current at various temperatures.

Ambient temperature °C	Factor
+105	0.3
+85	1.0
+65	1.4
+45	1.7
+25	2.0

When capacitors are operated at both a temperature and frequency other than +85°C and less than 20 kHz, respectively, both of the multiplying factors listed in tables II and III shall be used to obtain the permissible rms ripple current.

Under no conditions of temperature or frequency, regardless of the multipliers given above, should the rms ripple current exceed 20 amperes.

Insulation and grounding. These capacitors have an indeterminate insulation resistance between the cathode terminal and the case. Where a potential difference exists between the cathode terminal and the chassis or other metallic mounting surfaces (which are usually at ground potential), the capacitors should be mounted in such a manner as to insulate the container from such mounting surfaces.

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Operating life. These capacitors are expected to have a useful operating life in excess of 10 years when subjected to typical circuit environments and operating conditions. Life can be extended by derating with regard to environmental and electrical requirements.

Shelf life. Shelf life and reforming procedures shall be in accordance with MIL-STD-1311.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - ES

(Project 5910-1928)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19, 99

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